

1. Write a program (WAP) to display "Hello World" on console display. WAP to implement the following control characters:

'\n' is for new line, or you can use endl – cout<<endl<<"message";

'\t' is for tab ; '\a' is an alarm sound; '\r' is carriage return to go to the beginning of the current line

CODE :

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    cout << "Hello World" << endl;
```

```
    cout << "Hello\nWorld" << endl;
```

```
    cout << "Hello\tWorld" << endl;
```

```
    cout << "Beep sound coming.\a" << endl;
```

```
    cout << "Hello World\rHi" << endl;
```

```
    return 0;
```

```
}
```

```
! ) { . \ ! }  
Hello World  
Hello  
World  
Hello World  
Beep sound coming.  
Hillo World
```

2. Write a C++ program that will ask for a temperature in Celsius and display it in degree Fahrenheit. $F = 9C/5 + 32$

CODE :

```
#include<iostream>
```

```
int main() {
```

```
    double temp , result ;
```

```
    std::cout<<"Enter the temp : " ;
```

```
    std::cin>>temp ;
```

```
    result = (9*temp)/5 + 32 ;
```

```
    std::cout<<"the temperature in Fahrenhiet is : "<<result ;
```

```
    return 0 ;
```

```
}
```

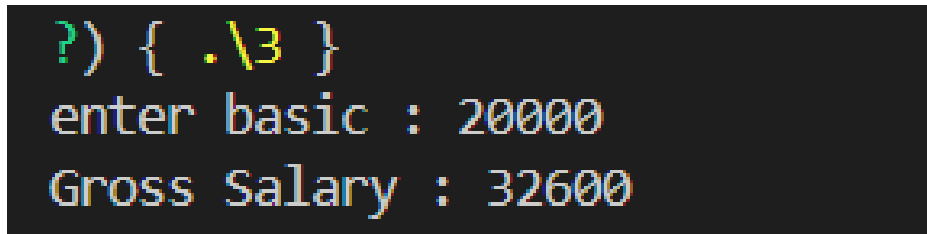
```
11e }  
Enter the temp : 100  
the temperature in Fahrenhiet is : 212
```

3. WAP to calculate gross salary of a person, where gross_salary = basic+da+ta+HRA and da is 46% of basic and ta is 8% of basic and HRA is 9% of basic pay.

CODE:

```
#include<iostream>  
  
int main() {  
    double basic , da , ta , hra , result ;  
  
    std::cout<<"enter basic : " ;  
    std::cin>>basic ;  
    da = 0.46*basic ;  
    ta = 0.08*basic ;  
    hra = 0.09*basic ;  
  
    result = basic + da + ta + hra ;  
  
    std::cout<<"Gross Salary : " << result ;
```

```
return 0 ;  
}
```

A screenshot of a C program's output. The first line shows a prompt character '?' followed by an opening curly brace '{', a backslash '\', and a closing curly brace '}', which likely represents a file path or a specific input/output redirection. The second line shows the user input 'enter basic : 20000'. The third line shows the program's output 'Gross Salary : 32600'.

```
? ) { .\3 }  
enter basic : 20000  
Gross Salary : 32600
```

4. A certain grade of steel is graded according to the following conditions:

- (i) Hardness must be greater than 50
- (ii) Carbon content must be less than 0.7
- (iii) Tensile strength must be greater than 5600

The grades are as follows:

Grade is 10 if all three conditions are met

Grade is 9 if all conditions (i) and (ii) are met

Grade is 8 if all conditions (ii) and (iii) are met

Grade is 7 if all conditions (i) and (iii) are met

Grade is 6 if only one condition is met

Grade is 5 if none of the conditions are met

Write a program, which will require the user to give values of hardness, carbon content and tensile strength of the steel under consideration and output the grade of the steel. (Solve using nested if-else, ladder if and else if)

CODE :

```
#include<iostream>
```

```
int main() {
```

```
double hardness , carbon_content , tensile_strength ;
```

```
std::cout<<"enter hardness of the steel : " ;
```

```
std::cin>>hardness ;
```

```
std::cout<<"enter carbon content of the steel : " ;
```

```
std::cin>>carbon_content ;
```

```
std::cout<<"enter tensile strength of the steel : " ;
```

```
std::cin>>tensile_strength ;
```

```
if(hardness>50 && carbon_content<0.7 && tensile_strength>5600) {
```

```
    std::cout<<"Grade 10" ;
```

```
}
```

```
else if(hardness>50 && carbon_content<0.7 )
```

```
{
```

```
    std::cout<<"Grade 9" ;
```

```
}
```

```
else if( carbon_content<0.7 &&tensile_strength>5600) {
```

```

        std::cout<<"Grade 8" ;
    }
    else if(hardness>50 && tensile_strength>5600) {
        std::cout<<"Grade 7" ;
    }
    else if(hardness>50 || carbon_content<0.7 || tensile_strength>5600) {
        std::cout<<"Grade 6" ;
    }

    else {
        std::cout<<"Grade 5" ;
    }

    return 0 ;
}

```

```

file }
enter hardness of the steel : 20
enter carbon content of the steel : 0.6
enter tensile strength of the steel : 3000
Grade 6

```

5. A library charges a fine for every book return late. For first 5 days the fine is 50 paisa, for 6-10 days fine is one rupee and above 10 days fine is

5 rupees. If you return the book after 30 days your membership will be cancelled. WAP to accept no. of days the member is late to return the book and display the fine or appropriate message.

Code :

```
#include<iostream>

int main() {

    int days , left ;

    double fine ;

    std::cout<<"enter number of days : " ;

    std::cin>>days ;

    if(days<=5) {

        fine = 0.5*days ;

        std::cout<<"fine : "<<fine ;

    }

    else if(days>=6 && days<=10) {

        left = days - 5 ;

        fine = 5*0.5 + left*1 ;

        std::cout<<"fine : "<<fine ;

    }

    else if(days>10 && days<=30) {

        left = days - 5 - 5 ;

        fine = left*5 + 5*0.5 + 5*1 ;

        std::cout<<"fine : "<<fine ;

    }

}
```

```

}
else {
    std::cout<<"membership is cancelled ....." ;
}

return 0;
}

```

```

116 }
enter number of days : 50
membership is cancelled .....

```

```

117 }
enter number of days : 13
fine : 22.5

```

6. WAP to demonstrate for, while, do-while (with all possible variations), like for loop can be demonstrated without giving initialization in for construct or without giving increment in for construct.

CODE :

```

#include<iostream>

int main() {

```



```
//variation of for loop

std::cout<<"For loop variation\n" ;

for(int i=0 ; i<10 ; i++) {

    std::cout<<i<<" " ;

}
```

```
std::cout<<"\n" ;

int i=0 ;

for(;i<10 ; i++) {

    std::cout<<i<<" " ;

}

std::cout<<"\n" ;

i=0 ;

for(; i<10 ; ) {

    std::cout<<i<<" " ;

    i++ ;

}
```

```
//variation for while loop

std::cout<<"\nwhile loop variation\n" ;

int j=0 ;

while(j<10) {

    std::cout<<j<<" " ;

    j++ ;

}

std::cout<<"\nDO While variations\n" ;

//variation in do while loop
```

```

int k = 0 ;

do{

    std::cout<<k<<" ";

    k++ ;

}while(k<10) ;

std::cout<<"\n" ;

//do while being excuted one time even thou the condition is wrong

k = -1 ;

do{

    std::cout<<k<<" ";

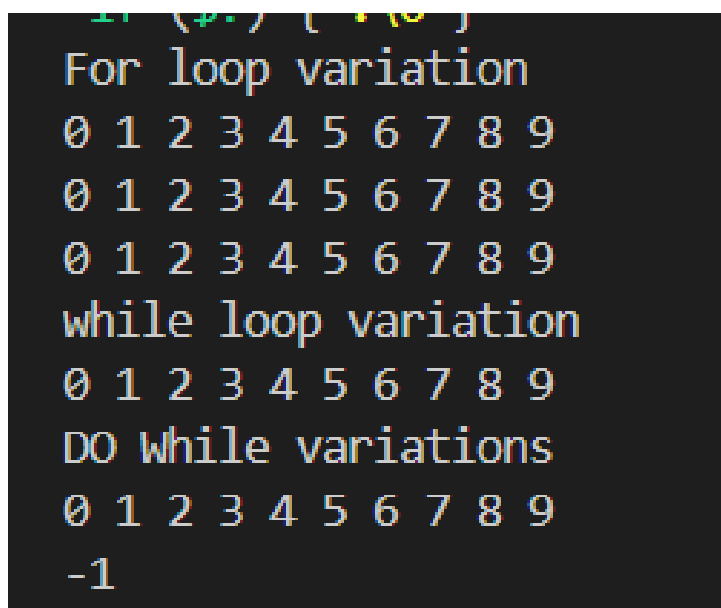
    k++ ;

}while(k>0) ;

return 0;

}

```



```

17 (4.7) (4.8)
For loop variation
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9
while loop variation
0 1 2 3 4 5 6 7 8 9
DO While variations
0 1 2 3 4 5 6 7 8 9
-1

```

7. Implement namespace in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.

CODE :

```
#include<iostream>

namespace fun1 {

    int x=19 ;
    int y=1 ;
    int add() {
        return x+y ;
    }
}

namespace fun2 {
    int x = 70 ;
    int y = 21 ;
    int add() {
        return x+y ;
    }
}

using namespace fun1 ;

int main() {

    std::cout<<fun2::x ;
    std::cout<<'\n'<<x ;
    std::cout<<'\n'<<fun1::add() ;
```

```
std::cout<<"\n"<<fun2::add() ;
```

```
}
```



```
70
19
20
91
```

8. Create a structure in C++ containing the details of Students as details below and a main function to execute the structure.

CODE :

```
#include<iostream>

struct student {

    //data members

    std::string name ;

    int roll_no ;

    std::string degree ;

    std::string hostel ;

    double currentcgpa ;

    //member function
```

```

void addDetails() {
    std::cout<<"enter the name : " ;
    std::cin>>name ;
    std::cout<<"enter roll : " ;
    std::cin>>roll_no ;
    std::cout<<"enter degree : " ;
    std::cin>>degree ;
    std::cout<<"enter hostel : " ;
    std::cin>>hostel ;
    std::cout<<"enter cgpa : " ;
    std::cin>>currentcgpa ;

}

void updateDetails() {
    std::cout<<"update degree : " ;
    std::cin>>degree ;

}

void updateCGPA() {
    std::cout<<"enter new cgpa : " ;
    std::cin>>currentcgpa ;
}

void updateResidenceInfo() {
    std::cout<<"enter new hostel info : " ;
    std::cin>>hostel ;
}

void displaydetails() {
    std::cout<<"\nNAME : " <<name ;

```

```
std::cout<<"\nROLL NO : "<<roll_no ;  
std::cout<<"\nDEGREE : "<<degree ;  
std::cout<<"\nHOSTEL : "<<hostel ;  
std::cout<<"\nCGPA : "<<currentcgpa ;  
  
}  
};
```

```
int main() {  
    student s1 ;  
    s1.addDetails() ;  
    s1.updateDetails() ;  
    s1.updateCGPA() ;  
    s1.updateResidenceInfo() ;  
    s1.displaydetails() ;  
  
    return 0 ;  
}
```

```
if ($?) { .\8 }
enter the name : anandi
enter roll : 5
enter degree : btech
enter hostel : pg
enter cgpa : 9.0
update degree : mtech
enter new cgpa : 8.9
enter new hostel info : house

NAME : anandi
ROLL NO : 5
DEGREE : mtech
HOSTEL : house
CGPA : 8.9
```

9. Differentiate between private and public access/scope. Perform the question no. 5 with structure having the data member in private scope. Students should be able to

a) differentiate between structure in C vs structure in C++

CODE :

```
#include<iostream>

struct fine {
    private :
        int days ;

        double fine ;
    public :
        int displayfine(int d) {
            days = d ;
            int left ;
            if(days<=5) {
                fine = 0.5*days ;
```

```

std::cout<<"fine : "<<fine ;

}

else if(days>=6 && days<=10) {

    left = days - 5 ;

    fine = 5*0.5 + left*1 ;

    std::cout<<"fine : "<<fine ;

}

else if(days>10 && days<=30) {

    left = days - 5 - 5 ;

    fine = left*5 + 5*0.5 + 5*1 ;

    std::cout<<"fine : "<<fine ;

}

else {

    std::cout<<"membership is cancelled ....." ;

}

}

};

int main() {

    fine l1 ;

    int day ;

    std::cout<<"enter how many days the book was late : " ;

    std::cin>>day ;

    l1.displayfine(day) ;

    return 0 ;

```



```
}
```

```
; if ($?) { g++ 9.cpp -o 9 } ; if ($?) { .\9 }  
enter how many days the book was late : 21  
fine : 62.5
```

```
; if ($?) { g++ 9.cpp -o 9 } ; if ($?) { .\9 }  
enter how many days the book was late : 21  
fine : 62.5
```

10. Create a code snippet that illustrates the following:

- a. Calling of private member functions inside public member function
- b. Access private member functions inside public member function

CODE :

```
#include<iostream>
```

```
struct number {
```

```
    int x = 3 ;
```

```
    int y = 2 ;
```

```
    private :
```

```
    int sum () {
```

```
        return x+y ;
```

```
    }
```

```
//calling of private member functions inside public member function .
```

```
    public :
```

```

void sum5() {
    int a = sum();
    a = a + 5;
    std::cout<<a;
}

};

int main() {
    number l;
    l.sum5();

}

```

we are calling public function and inside the public function we have private function call
10

11. An election is contested by five candidates. The candidates are numbered 1 to 5 and the voting is done by marking the candidate number on the ballot paper. Write a program to read the ballots and count the votes cast for each candidate using an array variable count. In case, a number read is outside the range 1 to 5, the ballot should be considered as a 'spoilt ballot', and the program should also count the number of spoilt ballots.

CODE :

```
#include<iostream>

int main() {

    int count , choice ;

    int candi[] = { 0 , 0 , 0 , 0 , 0 , 0 } ;

    std::cout<<"1) To vote\n" ;
    std::cout<<"2) To exit\n" ;

    do {

        std::cout<<"enter choice : " ;

        std::cin>>choice ;

        switch(choice) {

            case 1 :

                std::cout<<"enter the candidate u wanna vote : " ;

                std::cin>>count ;

                if(count>=1 && count<=5 ) {

                    candi[count-1]++ ;

                }

                else {

                    candi[5]++ ;

                }

            }

        }while(choice!=2) ;
```

```

for(int i=0 ; i<=5 ; i++) {

    if(i!=5){

        std::cout<<"candidate "<< i+1 <<" votes : " <<candi[i]<<"\n" ; }

        else {

            std::cout<<"spoilt vote : "<<candi[i]<<"\n" ;

        }

    }

}

return 0 ;

}

```

```

; if (p) { .\11 }
1) To vote
2) To exit
enter choice : 1
enter the candidate u wanna vote : 1
enter choice : 1
enter the candidate u wanna vote : 1
enter choice : 1
enter the candidate u wanna vote : 3
enter choice : 1
enter the candidate u wanna vote : 4
enter choice : 1
enter the candidate u wanna vote : 5
enter choice : 1
enter the candidate u wanna vote : 6
enter choice : 1
enter the candidate u wanna vote : 7
enter choice : 2
candidate 1 votes : 2
candidate 2 votes : 0
candidate 3 votes : 1
candidate 4 votes : 1
candidate 5 votes : 1
spoilt vote : 2

```

12. Write programs to evaluate the following functions to 0.0001% accuracy.

CODE :

```
#include <iostream>

#include <cmath>

using namespace std;

int main() {

    double x;

    cout << "Enter number (in radians): ";

    cin >> x;

    // ----- sin(x) using Taylor series -----

    int n = 1; // power

    int sign = 1;

    double term = x;    // first term = x^1 / 1!

    double sum = term;

    while (fabs(term) > 1e-6) {

        n += 2;           // next odd power

        sign = -sign;     // alternate sign

        term = sign * pow(x, n) / tgamma(n + 1);

        sum += term;

    }

    cout << "sin(x) = " << sum << endl;

    // ----- Series:  $\sum (1/n)^n$  -----
```

```

int n1 = 1;

double term1 = 1.0;  // first term

double sum1 = term1;

while (fabs(term1) > 1e-6) {
    n1++;
    term1 = pow(1.0 / n1, n1);
    sum1 += term1;
}

cout << "sum = " << sum1 << endl;


// ----- cos(x)

int n2 = 0;
int sign2 = 1;
double term2 = 1.0;
double sum2 = term2;

while (fabs(term2) > 1e-6) {
    n2 += 2;           // next even power
    sign2 = -sign2;
    term2 = sign2 * pow(x, n2) / tgamma(n2 + 1);
    sum2 += term2;
}

cout << "cos(x) = " << sum2 << endl;

return 0;
}

```

```

; if ($?) { g++ 12.cpp -o 12 } ; if (
Enter number (in radians): 1
sin(x) = 0.841471
sum = 1.29129
cos(x) = 0.540302

```

13. An electricity board charges the following rates to domestic users to discourage large consumption of energy:

For the first 100 units

60P per unit

For next 200 units

80P per unit

Beyond 300 units

90P per unit

All users are charged a minimum of Rs. 50.00. If the total amount is more than Rs.

300.00 then an additional surcharge of 15% is added. Write a program to read the

names of users and number of units consumed and print out the charges with names

CODE :

```
#include<iostream>
```

```
int main() {
```

```
    std::string name ;
```

```
    int units , left ;
```

```
    double bill ;
```

```
std::cout<<"enter your name : ";
```

```
std::cin>>name ;
```

```
std::cout<<"enter number of units consumed : " ;
```

```
std::cin>>units ;
```

```
if(units<=100) {
```

```
    bill = units*0.6 ;
```

```
}
```

```
else if(units<=300) {
```

```
    left = units - 100 ;
```

```
    bill = 100 * 0.6 + left*0.8 ;
```

```
}
```

```
else {
```

```
    left = units - 100 - 200 ;
```

```
    bill = 100*0.6 + 200*0.8 + left*0.9 ;
```

```
}
```

```
if(bill<=50) {
```

```
    std::cout<<"Name : "<< name<<" charge : : 50" ;
```

```
}
```



```
else if (bill > 50 && bill <= 300) {  
    std::cout << "Name : " << name << " charge : : " << bill ;  
}  
else {  
    bill = bill + 0.15 * bill ;  
    std::cout << "Name : " << name << " charge : : " << bill ;  
}  
  
return 0 ;  
  
}
```

```
enter your name : lara  
enter number of units consumed : 40  
Name : lara charge : : 50
```